[INCH-POUND]
A-A-59440/3A
23 May 2005
SUPERSEDING
A-A-59440/3
13 August 1999

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

# COCK, DRAIN, SPRING KEY, BRASS, 50 P.S.I., WATER, OIL, GAS (WOG), NPTF-1 THREADS

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

The complete requirements for procuring the cock, drain, spring key, brass, 50 psi, water, oil, gas (WOG), NPTF-1 threads described herein shall consist of this document and the issue in effect of CID A-A-59440.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

| AA59440 | /3 | - | 011 |
|---|---|---|---|
| CID number | Applicable CID sheet (two digits) | | Dash number (see figure 1) |

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Cock, drain, spring key, brass, 50 psi, water, oil, gas (WOG), NPTF-1 threads supplied to this CID shall be as specified herein (see figure 1).

Sizes. Nominal sizes of 1/8 inch through 3/8 inch with NPTF-1 threads.

Intended application. These items are for use in liquid or gas systems where the pressure does not exceed 50 psi (3.45 bar) for water, oil and gas, (WOG) at 150°F (65.6°C) and the liquid is compatible with brass. Spring material is phosphor bronze, carbon steel, or corrosion resistant stainless steel (CRES) (see figure 1).

Permissible variations in dimensions. The table in figure 1 lists the permissible variation in dimensions for each size. The external dimensions of the cocks should not limit procurement unless the end item applications restrict these dimensions.

Optional designs. Figure 1 is for identification purposes only and is not intended to restrict designs or shapes not dimensioned.

Threads. Threads are dryseal NPTF-1 in accordance with CID A-A-59440.



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Material.

Material body, handle, stem, and screw. Body, handle, stem, and screw shall be brass in accordance with the following commercial standards:

| | | | |
|---|---|---|---|
| ASTM B16 | ASTM B121 | ASTM B505 | SAE AMS 4842 |
| ASTM B22 | ASTM B124 | ASTM B584 | SAE AMS 4845 |
| ASTM B30 | ASTM B148 | ASTM B633 | SAE AMS 4855 |
| ASTM B36 | ASTM B176 | ASTM B763 | SAE AMS 4860 |
| ASTM B61 | ASTM B271 | ASTM B770 | SAE AMS 4862 |
| ASTM B62 | ASTM B369 | ASTM B806 | SAE AMS 4890 |
| | | ASTM B427 | |

Spring material. Spring material as specified by the corresponding dash number in figure 1 and shall be in accordance with one of the following:

a. Phosphor bronze in accordance with ASTM B134, ASTM B159, ASTM B206 alloy number 510.
b. Carbon steel in accordance with SAE AMS-QQ-W-428.
c. Corrosion resisting steel (CRES) in accordance with ASTM A313, composition 302, 304, 305, 316, condition B spring temper.

Spring finish.

a. Phosphor bronze: None.
b. CRES: None.
c. Carbon steel:
   (1) Cadmium in accordance with SAE AMS-QQ-P-416, type II, class 2.
   (2) Zinc in accordance with ASTM B633, type II, 12.7 µm (50 microinches) minimum.

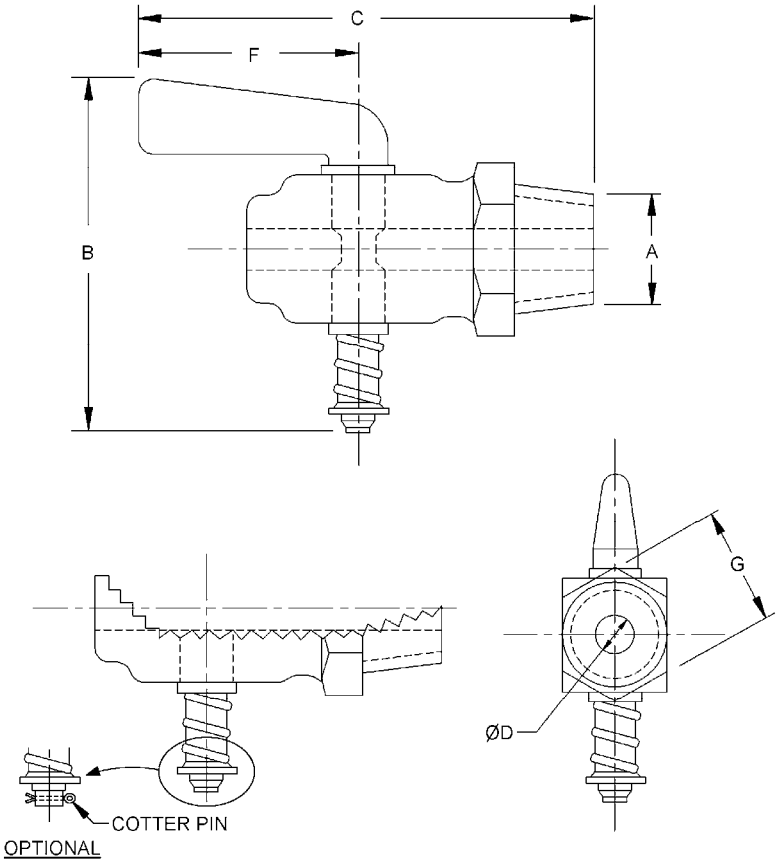
Phosphor bronze springs. Phosphor bronze springs when specified are used to avoid possible spark and explosion.

Test pressures.

a. Shell 80 psig (5.12 bar)
b. Seat 10 psig (0.69 bar)



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| Dash Number | Spring Material | A Pipe thread NPTF-1 | B Max (mm) | C Max (mm) | D dia Min (mm) | F Max (mm) | G across flats ±.015 (0.38 mm) 1/ |
|---|---|---|---|---|---|---|---|
| 011 | Phosphor Bronze | 1/8-27 | 1.75 (44.7) | 1.91 (48.5) | .16 (4.1) | 1.00 (25.4) | .44 or .50 (11.2 or 12.7) |
| 012 | Phosphor Bronze | 1/4-18 | 1.88 (47.6) | 2.47 (62.7) | .19 (4.8) | 1.22 (31.0) | .56 or .62 (14.2 or 15.7) |
| 013 | Phosphor Bronze | 3/8-18 | 2.75 (69.9) | 2.97 (75.4) | .23 (5.8) | 1.50 (38.1) | .69, .75 or .88 (17.5, 19.1, or 22.4) |
| 015 2/ | CRES | 1/4-18 | 1.88 (47.6) | 2.47 (62.7) | .19 (4.8) | 1.22 (31.0) | .56 or .26 (14.2 or 15.7) |
| 016 2/ | Carbon Steel (cadmium plated) | | | | | | |
| 017 2/ | Carbon Steel (zinc plated) | | | | | | |

1/ Hex flat measurement. Hexagonal hex flats can be a standard wrench pad.
2/ Caution: If other than phosphor bronze spring is used be aware of possible spark and explosion.

FIGURE 1. Cock, Drain, with Hose Bib.



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NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.010 inch (0.25 mm).
4. Diameter of hole shall be adequate to allow drainage through the valve.

FIGURE 1. Cock, Drain, with Hose Bib - Continued.

NOTE.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See classification information for PIN format example.

Source of documents.

COMMERCIAL ITEM DESCRIPTIONS

| | | |
|---|---|---|
| A-A-59440 | - | Cocks, Drain, Cocks, Plug, and Cocks, Shutoff, Screw Stem; Brass, NPT and NPTF-1 Threads, General Requirements For |

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)



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Other Publications

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM A313 | - | Wire Stainless Steel Spring |
| ASTM B16 | - | Rod, Brass, Free-Cutting, Bar and Shapes for use in Screw Machines |
| ASTM B22 | - | Castings, Bronze, for Bridges and Turntables |
| ASTM B30 | - | Copper Alloys in Ingot Form |
| ASTM B36 | - | Plate, Brass, Sheet, Strip, and Rolled Bar |
| ASTM B61 | - | Castings, Steam or Valve Bronze |
| ASTM B62 | - | Metal Castings, Composition Bronze or Ounce |
| ASTM B121 | - | Plate, Leaded Brass, Sheet, Strip, and Rolled Bar |
| ASTM B124 | - | Copper and Copper Alloy Forging Rod, Bar, and Shapes |
| ASTM B134 | - | Brass Wire |
| ASTM B148 | - | Aluminum-Bronze Sand Castings |
| ASTM B159 | - | Wire, Phosphor Bronze, Standard Specification for |
| ASTM B176 | - | Die Castings, Copper-Alloy |
| ASTM B206 | - | Copper-Nickel-Zinc (Nickel Silver) Wire and Copper-Nickel Alloy Wire |
| ASTM B271 | - | Copper-Base Alloy Centrifugal Castings |
| ASTM B369 | - | Copper-Nickel Alloy Castings |
| ASTM B427 | - | Castings, Gear Bronze Alloy |
| ASTM B505 | - | Castings, Copper Alloy, Continuous |
| ASTM B584 | - | Castings, Copper Alloy Sand, for General Applications |
| ASTM B633 | - | Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel |
| ASTM B763 | - | Copper Alloy Sand Castings for Valve Application |
| ASTM B770 | - | Copper-Beryllium Alloy Sand Castings for General Applications |
| ASTM B806 | - | Copper Alloy Permanent Mold Castings for General Applications |

(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, Pennsylvania, USA 19428-2959.)

SAE INTERNATIONAL

| | | |
|---|---|---|
| SAE AMS-QQ-P-416 | - | Plating, Cadmium (Electrodeposited) |
| SAE AMS-QQ-W-428 | - | Wire, Steel, Carbon (High Carbon, Round, for Mechanical Springs, General Purpose) |
| SAE AMS4842 | - | Castings, Leaded Bronze, Sand and Centrifugal 80CU - 10SN - 9.5PB as Cast |
| SAE AMS4845 | - | Tin Bronze Castings, Sand and Centrifugal 87.5CU - 10SN - 2ZN as Cast |
| SAE AMS4855 | - | Leaded Red Brass, Sand and Centrifugal Castings 85CU - 5.0SN - 5.0PB - 5.0ZN as Cast |
| SAE AMS4860 | - | Manganese Bronze, Sand and Centrifugal Castings 58CU - 39ZN - 1.2FE - 1.0AL - 0.80MN as Cast |
| SAE AMS4862 | - | Manganese Bronze, Sand and Centrifugal Castings 63CU - 24ZN - 6.2AL - 3.8MN - 3.0FE High Strength, as Cast |
| SAE AMS4890 | - | Copper-Beryllium Alloy Castings 97CU - 2.1BE - 0.52CO - 0.28SI Solution Heat Treated (TBOO) |

(Copies of these documents are available from http://www.sae.org/ or SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)



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Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

| MFR's CAGE | MFR's name and address |
|---|---|
| 79470 | Eaton Corp<br>14615 Loneoak Rd.<br>Edenperry, MN 55344<br>952-294-2032/888-258-0222<br>jeffenoe@eaton.com |
| 93061 | Parker-Hannifin Corp<br>Brass Products Division<br>300 Parker Drive<br>Otsego, MI 49708-1472<br>260-636-2104<br>dwsanders@Parker.com |
| 20969 | Essex Brass Corp<br>23500 Pinewood<br>Warren, MI 48091-3122<br>586-757-8200 |

Military Standard (MS) replacement data. This CID replaces the following Military Standard as shown in table I.

TABLE I. MS supersession data.

| CID PIN | Superseded MS |
|---|---|
| AA559440/3-011 | MS35784-1 |
| AA559440/3-012 | MS35784-2 |
| AA559440/3-013 | MS35784-3 |
| AA559440/3-016 1/ | MS35784-4 CRES |
| AA559440/3-017 1/ | MS35784-4 Carbon steel (cadmium plated) |
| AA559440/3-018 1/ | MS35784-4 Carbon steel (zinc plated) |

1/ These parts were originally listed in MS35784 as a single part type. Due to special handling/disposal and corrosion resistance requirements the parts have been separated in this CID.



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Part number (P/N) supersession data. These CID P/N's supersede the following MFR's P/Ns as shown in table II. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

| CID PIN | MFR's CAGE | MFG's P/N 1/ |
|---|---|---|
| AA59440/3-011 | 20969<br>70411 | AC63L<br>30CJ |
| AA59440/3-012 | 20969 | AC64L |
| AA59440/3-013 | 20969 | AC14 |
| AA59440/3-016 | 20969 | N/A |
| AA59440/3-017 | 20969 | N/A |
| AA59440/3-018 | 20969 | N/A |

1/ The manufacturer's PIN shall not be used for procurement to the requirements of this CID. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements see marking paragraph.

National stock number (NSN). The following is a list of NSN's assigned which correspond to this CID. The list in table III is for information only and may not be indicative of all possible NSN's associated with the CID. For up to date information on assigned NSN's, please contact the aforementioned DSCC office (see A-A-59440).

TABLE III. NSN's

| Dash number (see figure 1) AA59440/3- | NSN |
|---|---|
| 011 | 4820-00-555-9761 |
| 013 | 4820-00-272-3346 |

Government users. To acquire information on obtaining these cocks from the Government inventory system, contact Defense Supply Center, Columbus, ATTN: DSCC-NAB, P.O. Box 3990, Columbus, OH 43218-3990, or telephone (614) 692-2271 or (614) 692-3191.

Marginal notations. Marginal notations are not used in this revision to identify changes due to the extent of the changes.



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MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99
DLA - CC

Review activities:
Army - AT, AV, CR4
Navy - MC

CIVIL AGENCY AND COORDINATING ACTIVITIES:

GSA - FSS

Preparing activity:

DLA - CC

Project 4820-0853-000

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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